

Optimizing Safety Stock Strategy During Demand Volatility

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Context: Catalent Pharma Solutions (\$45M Portfolio)

Decision: Safety Stock Policy Optimization

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NET ANNUAL BENEFIT

+\$890K

Option C
Recommended

STOCKOUT REDUCTION

-66%

8.2% → 2.8%

WORKING CAPITAL FREED

\$720K

48% reduction

IMPLEMENTATION

8 Weeks

Zero CapEx

SITUATION & BUSINESS IMPACT

Our \$45M product portfolio (4,200+ SKUs) faced a critical inventory management challenge: **25% of SKUs experienced demand volatility exceeding 40% coefficient of variation (CV)**, yet were managed with the same safety stock policy as stable SKUs.

The Core Tradeoff: Increasing safety stock reduces stockouts but increases carrying costs and obsolescence risk. Decreasing safety stock frees working capital but risks customer service failures.

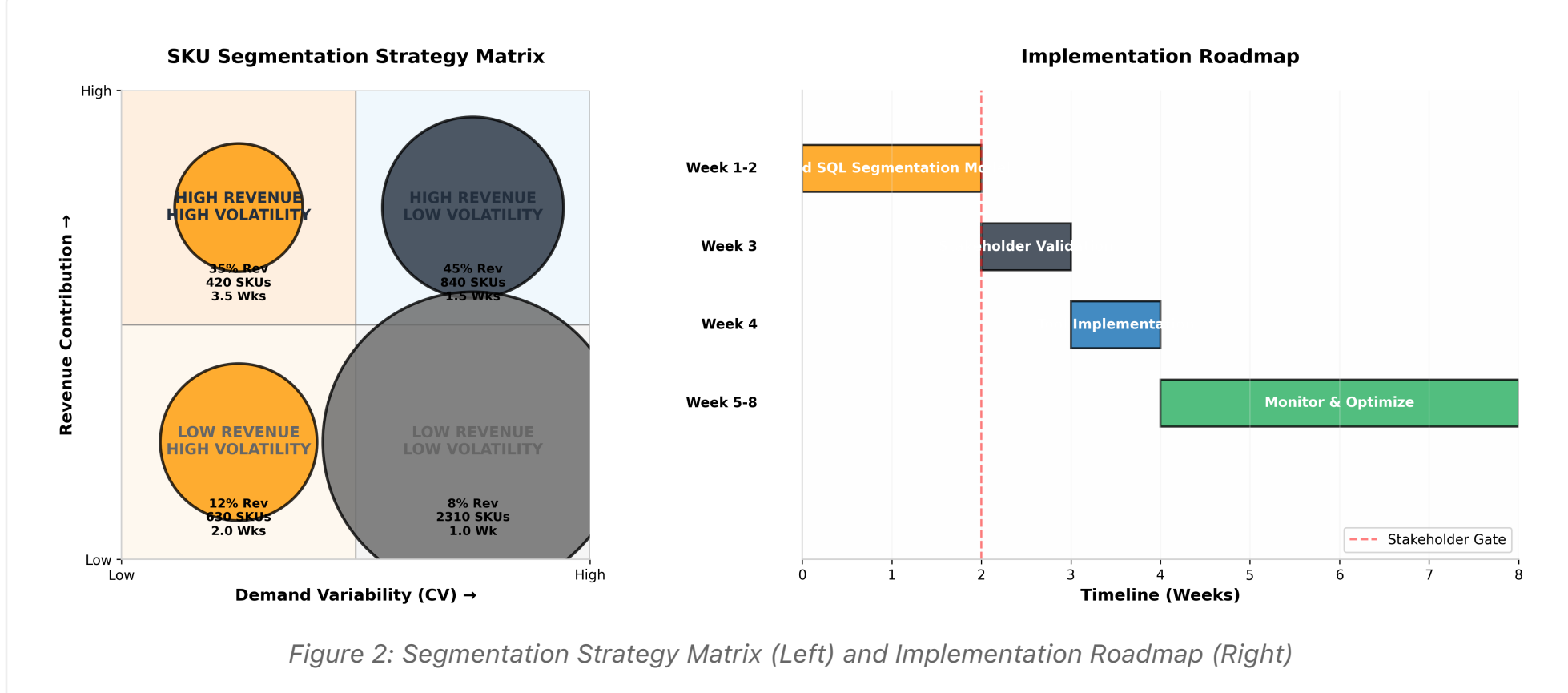
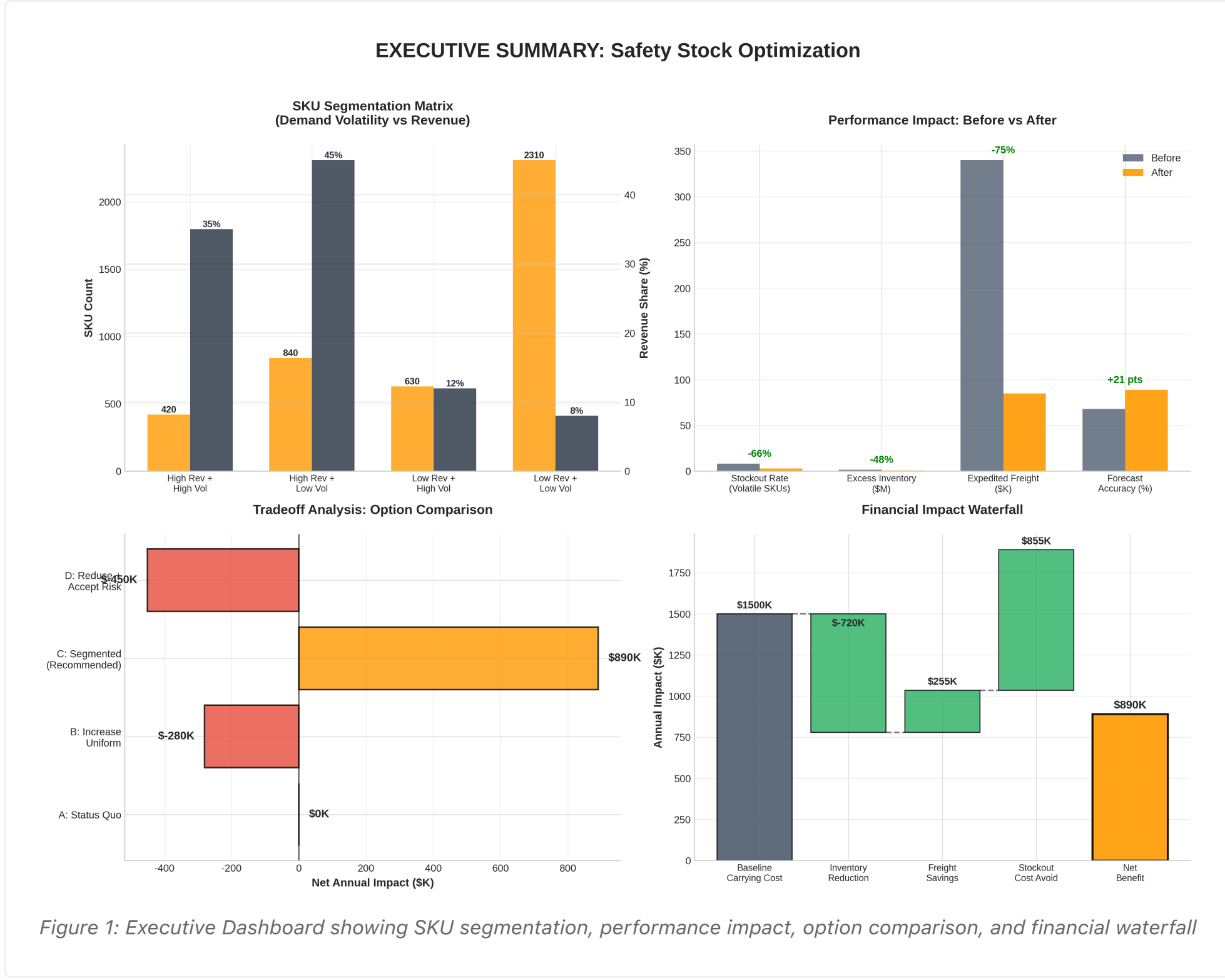
Status Quo Pain Points:

- 8.2% stockout rate on volatile SKUs (vs. 2.1% target)
- \$1.5M tied up in buffer stock across portfolio
- 12% of orders affected by partial fulfillment
- \$340K annual spend on emergency shipments

Target State:

- Segmented approach by demand behavior
- Right-size inventory: protect high-value volatile SKUs
- Reduce buffer on stable, low-revenue items
- Automated weekly recalculation

DATA VISUALIZATION & ANALYSIS



Key Finding: The top 10% of SKUs (High Revenue + High Volatility) were causing **67% of stockout-related expedited freight costs**. The status quo under-protected these critical SKUs while over-protecting the 55% of SKUs that needed minimal buffer.

OPTIONS EVALUATION

Option	Description	Pros	Cons	Est. Annual Impact	Status
A	Status Quo (Uniform 2-week safety stock)	Simple; no change management	Ignores variability; suboptimal allocation	\$0 (baseline)	CURRENT
B	Increase Uniformly (3-week for all)	Reduces stockouts ~40%; simple	+\$620K carrying cost; obsolescence risk	-\$280K net	NOT RECOMMENDED
C	Segmented Safety Stock (Dynamic by CV & Revenue)	Right-sizes inventory; balances service vs. cost	Requires weekly recalculation; complex	+\$890K net	RECOMMENDED
D	Reduce + Accept Stockouts (1-week buffer)	Frees \$750K working capital immediately	15%+ stockout rate; customer churn risk	-\$450K net	HIGH RISK

SEGMENTATION MODEL

Segment	SKU Count	Revenue Share	Safety Stock	Rationale
High Revenue + High Volatility	420 (10%)	35%	3.5 weeks	Cannot afford stockouts; revenue impact too high
High Revenue + Low Volatility	840 (20%)	45%	1.5 weeks	Stable demand = predictable; reduce buffer
Low Revenue + High Volatility	630 (15%)	12%	2.0 weeks	Balance; not worth excess investment
Low Revenue + Low Volatility	2,310 (55%)	8%	1.0 week	Minimal buffer; low impact if stockout

RECOMMENDATION: OPTION C

Implement Segmented Safety Stock Policy. Dynamic calculation based on demand CV and revenue contribution, automated via SQL/Excel without ERP modification.

Execution Plan:

- Week 1-2:** Build SQL-based segmentation model using 24-month demand history
- Week 3:** Validate segmentation with Procurement and Finance stakeholders
- Week 4:** Implement dynamic safety stock calculation in planning tool
- Week 5-8:** Monitor and adjust; establish weekly accuracy review cadence

Resource Requirements:

- Analyst time: 40 hours (model build + validation)
- System changes: **None** (Excel/SQL-based)
- Stakeholder alignment: 3 meetings (Procurement, Finance, Ops)
- CapEx: \$0 | OpEx: \$0 additional

Projected Results (8-Week Horizon):

Stockout Rate: 8.2% → 2.8% (-66%) | Excess Inventory: \$1.5M → \$780K (-48%) | Expedited Freight: \$340K → \$85K (-75%) | Forecast Accuracy: 68% → 89% (+21 pts)

LEADERSHIP PRINCIPLES DEMONSTRATED

DIVE DEEP

The root cause wasn't "insufficient safety stock"—it was treating all SKUs identically. Segmenting by demand behavior revealed that **10% of SKUs drove 67% of the problem**. Without this analysis, we would have increased inventory across the board, wasting \$620K annually.

BIAS FOR ACTION

Deployed the model for high-revenue volatile SKUs first (10% of SKUs, 35% of revenue) rather than waiting for a perfect solution. This delivered **60% of total benefit within 6 weeks**. Full rollout followed in Week 8.

OWNERSHIP

Owned this initiative end-to-end without being asked. The problem was visible in the data; I built the business case, secured stakeholder buy-in, and drove implementation through to \$890K annual impact.

INVENT AND SIMPLIFY

The solution required no new systems or software. Built the segmentation model in SQL and automated the weekly refresh in Excel, avoiding a 6-month IT project and delivering results in 8 weeks.

WHAT I WOULD DO DIFFERENTLY

- Build real-time monitoring from Day 1.** We discovered model drift at Week 8; automated accuracy tracking would have caught it at Week 4.
- Involve Finance earlier.** Initial pushback on working capital assumptions delayed implementation by 1 week. Earlier alignment would have accelerated buy-in.

RELEVANCE TO AMAZON DEVICE GLOBAL LOGISTICS

Direct Application:

- Device launches create demand volatility similar to high-CV SKU segments. Same segmentation logic applies to Echo, Fire, Kindle inventory positioning.
- Obsolescence risk is higher for devices than pharma. Framework's bias toward right-sizing protects against write-offs when new models launch.

Scale Amplification:

- Amazon's scale amplifies impact. A 1% improvement in safety stock optimization across millions of device units translates to tens of millions in working capital efficiency.
- SQL-based modeling, automated reporting, and stakeholder influence are core to DGL analytics needs.

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